



Wildfire narratives: Identifying and characterizing multiple understandings of western wildfire challenges

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ABSTRACT

Western wildfires present a complex sustainability challenge characterized by more severe fires and escalating risks. To mitigate western wildfire risks, collaborative management practices need to transform the processes involved in knowledge production, seizing the opportunities and overcoming obstacles associated with actors' multiple understandings. Knowledge co-production represents an increasingly referenced process for bringing together diverse actors, including scientists from different disciplines and non-scientists, to construct place-based and action-oriented knowledge. While knowledge co-production scholarship emphasizes the importance of recognizing and legitimizing multiple understandings, processes for attending to multiple understandings lack a systematic exploratory method for identifying and characterizing what those understandings are. We employed a narrative analysis to empirically investigate multiple understandings within the context of western wildfire challenges. Based on sixty semi-structured interviews with influential actors, we identified nine social narratives that capture distinctions in the connections actors make between the causes, consequences, and solutions to wildfire challenges, in the spatial and temporal scale they emphasize, in the way they frame the challenge, and in the language they use. We also found differences in how actors demarcate social narratives' credibility, legitimacy, and saliency. Our research suggests that analyzing social narratives fills an essential gap in practice by providing a pragmatic exploratory process for identifying and characterizing actors' multiple understandings.

1. Introduction

Around the globe, wildfires present a complex sustainability challenge characterized by more severe fires and escalating risks (Bowman et al., 2020; Burke et al., 2021; Dunn et al., 2020; Safford et al., 2022). Addressing these challenges necessitates management decisions that function across jurisdictions, agencies, and ownerships (Quinn-Davidson and Varner, 2012; Yung et al., 2022). For example, the Collaborative Forest and the Joint Chiefs Landscape Restoration Programs are national programs that inform landscape-scale forest restoration, including wildfire risk management, through collaborative decision-making (Schultz et al., 2012). Regional and local efforts have sprung up across the West, including numerous interagency partnerships, like the West Region Wildfire Council and Firewise councils like FireWise of Southwest Colorado, that educate, plan, and mitigate potential wildfire threats (Brenkert-Smith et al., 2017). However, despite both the growing saliency and significant investments in risk management collaborations, facilitating change in western wildfire management systems has been limited (Glenn et al., 2022; Tedim et al., 2021).

Several authors and policy advocates suggest that a critical limitation to the success of cross-boundary wildfire risk management is not technical or financial but rather the task of working across diverse ways of understanding the challenge (Paveglio, 2021; Yung et al., 2022).

Collaborative risk management that addresses complex sustainability challenges, including wildfires, flooding, forest pests, and watershed health is frustrated by irreducible uncertainty, values in dispute, high stakes, and urgency (Funtowicz and Ravetz, 1993; Wyborn et al., 2019). These challenges harbor multiple epistemically sound scientific perspectives, suggesting a need for processes that attend to multiple scientific viewpoints (Chambers et al., 2021). Beyond scientific knowledge, other ways of knowing hold lessons for responding to change and nurturing diversity (Folke et al., 2003). Indigenous, experiential, local, traditional, and professional ways of knowing reflect approaches to understanding and engaging with the world that emphasize different epistemologies, knowledge sources, ethical considerations, and communication (Hakkarainen et al., 2022; Klenk and Meehan, 2015; Norström et al., 2020). At the same time, differences in understandings have proven to be a formidable barrier to advancing risk management

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practices, emphasizing the importance of finding common ground or shared understandings (Davis et al., 2021; Morehouse and Sonnett, 2010; Moritz et al., 2018; Paveglio, 2021). Management practices aiming to mitigate western wildfire risks necessitate more effective processes for producing knowledge to inform management decisions, both seizing opportunities and overcoming obstacles associated with multiple understandings (Glenn et al., 2022; Paveglio, 2021).

Environmental governance is turning away from traditional knowledge production modes, where research is intentionally separated from practice (Jagannathan et al., 2020; Nowotny et al., 2001) and towards knowledge co-production (Chambers et al., 2021; Lemos et al., 2018; Zurba et al., 2022). This collaborative and interactive strategy brings together diverse actors, including scientists from different disciplines and non-scientists, to construct place-based and action-oriented knowledge (Armitage et al., 2011; Jagannathan et al., 2020; Norström et al., 2020). A scholarly principle of knowledge co-production is embracing a range of perspectives on an issue, considering the inevitable partiality and positioning of all knowledge (Norström et al., 2020). Intentionally involving diverse actors in knowledge production processes implies accepting that there can be multiple ways of interpreting or making sense of a problem (Brugnach and Ingram, 2012). The contribution of diverse understandings is theorized to create knowledge that is not only of high scientific quality but also socially robust (Turnhout et al., 2020). However, recent evaluations of knowledge co-production suggest that its theoretical aspirations may not match outcomes in practice (Jagannathan et al., 2020; Turnhout et al., 2020).

While knowledge co-production scholarship emphasizes recognizing and legitimizing multiple understandings, processes for attending to multiple understandings may be limited. Research suggests that processes may not represent a broad enough spectrum of understandings (Brugnach and Ingram, 2012; Chakraborty et al., 2022; Jagannathan et al., 2020; Polk, 2015), accurately distinguish between understandings (Chambers et al., 2022; Hakkarainen et al., 2022), or address discursive power, i.e., the ability to control the discourse or how a situation is collectively talked about (Purdy and Jones, 2012). These limitations may perpetuate power inequities (Gray et al., 2022), marginalize voices (Purdy and Jones, 2012; Tengö et al., 2014), erode trust (Ansell et al., 2020), destabilize agreements (Cosens, 2013), and make conflicts more intractable (Dewulf et al., 2009; Lewicki et al., 2003). To recognize actors' multiple understandings of complex sustainability challenges, practitioners require a systematic exploratory method for identifying and characterizing what those understandings are (Chakraborty et al., 2022; Proctor, 2020; Tengö et al., 2014). To legitimize multiple understandings, collaborative practices need a framework for revealing actors' assumptions about what counts as knowledge, i.e., perceptions about legitimacy, credibility, and saliency (Brugnach and Ingram, 2012; Cash et al., 2002; Daly, 2016).

This paper reports on a systematic exploratory method for identifying and characterizing influential actors' multiple understandings of increasingly severe wildfires in the forested western United States, hereafter referred to as "western wildfire challenges." Influential actors represent recognized experts from multiple disciplines and actors with informed understandings outside of academia who are shaping public discourse and thinking around western wildfire challenges. We interviewed sixty actors to elicit their understandings and conducted a narrative analysis to identify nine social narratives depicting wildfire challenges. Social narratives are common stories multiplied in the social area, i.e., publicly told and retold, to depict shared experiences and interpretations of what is happening and what should be done (Rawluk et al., 2020; Shenhav, 2015). The research was guided by the following questions, as informed by the literature review: First, to identify a representative breadth of understandings of the challenge, we asked - what are the social narratives of western wildfire challenges? Second, to explicitly characterize differences in understandings, we asked - what are the important distinctions between influential actors' social narratives? Third, to address discursive power by legitimizing multiple

understandings, we asked - how do influential actors demarcate the boundaries of credible, salient, and legitimate social narratives regarding western wildfire challenges?

2. Processes for recognizing and legitimizing multiple understandings

Recognizing and legitimizing actors' understandings remains a core challenge in knowledge co-production processes (Brugnach and Ingram, 2012; Horcea-Milcu et al., 2022). Processes for representation may be inadequate in capturing a diversity of values, perspectives, and interests (Turnhout et al., 2020). Furthermore, processes for deliberation may be insufficient in ensuring that diverse understandings are included in a legitimate way (Turnhout et al., 2020). The initial exploratory stage of figuring out the different understandings is very resource-intensive but largely underfunded (Horcea-Milcu et al., 2022; Page et al., 2016; Steger et al., 2021). Existing processes often lack guidelines and are poorly documented (Chakraborty et al., 2022; Horcea-Milcu et al., 2022; Reed and Abernethy, 2018; Wyborn et al., 2019). Foremost, there is substantial conceptual ambiguity in what scholarship suggests is captured by understandings (Bremer and Meisch, 2017). For example, publications reference integrating diverse perspectives, evidence, knowledge systems, ways of knowing, and epistemologies (Bandola-Gill et al., 2023; Chapman and Schott, 2020; Kalafatis et al., 2015; Klenk and Meehan, 2015; Roux et al., 2017; Zurba et al., 2022). This conceptual ambiguity may exacerbate procedural limitations shaping the design and evaluation of knowledge co-production processes (Fazey et al., 2014).

This paper uses the term understandings to refer to how actors construct meaning, interpret, or make sense of knowledge. We do not attempt to distinguish knowledge from different experiences, cultures, ways of knowing, disciplines, evidence, or values, but rather how these ideas come together to form actors' views about the problem and how it ought to be addressed. We adopt a constructivist-interpretive stance that individuals actively construct and interpret their understandings through personal experiences, social interactions, and mental processes (Creswell, 2007; Fazey et al., 2014; McCarthy, 2006; Van Der Walt, 2020). This conceptualization defines understandings as plural, partial, and positioned (Albrechts, 2013; Bremer and Meisch, 2017; Chakraborty et al., 2022; Chambers et al., 2022; Gunderson and Holling, 2002; Miller et al., 2008; Schuttenberg and Guth, 2015; Williams, 2014). As such, each actor's perceived reality of the right or most appropriate interpretations of western wildfire challenges is neither right nor wrong but rather frames the situation or emphasizes different elements of the challenge in different ways.

2.1. Identifying a representative breadth of understandings

Ideally, co-produced knowledge incorporates a breadth of understandings mirroring the complexity and diversity of the problem (Gray, 1989; Knapp et al., 2019; Roux et al., 2017). Capturing a representative breadth of multiple understandings supports more comprehensive knowledge (Mascarenhas et al., 2021; Susskind, 2010) and just and inclusive practices (Wyborn, 2015). A representative breadth goes beyond majority views to capture the diversity of ways actors think about the problem (Polk, 2015). Breadth is often determined by calling for actors from diverse affiliations or interests (Reed and Abernethy, 2018). However, it is unclear whether these diversity attributes also capture multiple understandings (Chakraborty et al., 2022; Muñoz-Erickson, 2014). Recent evidence suggests that differences in sectors and interests (e.g., tribal government, private sector, innovation) may not adequately indicate different perspectives (Brouwers et al., 2022). Breadth can be further limited by a narrow focus on scientific knowledge (i.e., processes guided by science-based decision-making), delegitimizing other voices (Turnhout et al., 2020). For example, practices that advance a particular scientific understanding through outreach and education (Brenkert-Smith et al., 2017; McCaffrey, 2004;

Yung et al., 2022) can marginalize the legitimacy of other views.

2.2. Explicitly characterizing differences in understandings

A second fundamental challenge in knowledge co-production processes lies in characterizing the diverse dimensions along which actors' understandings differ. Characterizing differences can inform process design and influence project success (Fazey et al., 2014; Rathwell et al., 2015). Differences in understandings can foster creativity and flexibility (i.e., are complementary), but differences can also cause miscommunication, disagreements, and conflicts (i.e., are conflicting) (Mannix and Neale, 2005; Salazar et al., 2012). Finding parallels, for example, between indigenous and scientific knowledge systems can enhance the legitimacy and credibility of both (Rathwell et al., 2015; Tengö et al., 2014). Alternatively, turning to evidence when differences stem from deeper conflicts may unintentionally reinforce existing tensions and inhibit collective action, making challenges more intractable (Dewulf et al., 2009; Lewicki et al., 2003). In some cases, differences should be recognized but not reconciled. For example, scholars have questioned whether the integration (i.e., adding together) of diverse perspectives may coerce indigenous or local knowledge to align with standards and values associated with western scientific understandings (Klenk and Meehan, 2015; Tengö et al., 2014).

2.3. Addressing discursive power by legitimizing multiple understandings

A third critical weakness in the success of knowledge co-production is the capacity to change societal power relations, particularly what understandings are privileged in decision-making (Wyborn et al., 2019). On par with power over territories and resources is discursive power, (Purdy and Jones, 2012). Actors with shared experiences, disciplines, cultures, or personal beliefs may have access to the same evidence, hold similar perspectives, or frame issues similarly, resulting in shared understandings. Discursive power emerges when shared understandings become commonly accepted and reinforced within institutions as the right view of a policy problem (Stone, 2012). Discursive power can manifest in knowledge co-production processes as power over defining the problem and project scope and determining what evidence is credible, legitimate, and salient (Cash et al., 2002; Goldstein and Butler, 2010; Muñoz-Erickson, 2014). Paradoxically, while knowledge co-production scholarship emphasizes legitimizing multiple understandings, different ways of talking about the situation are treated differently in practice. The power to sculpt the dominant narrative is not shared evenly across society (Brenkert-Smith et al., 2017). Policy elites, or actors from government, large NGOs, and the sciences with more time and resources tend to command discursive legitimacy, articulating contributions that align with the dominant narrative (Brenkert-Smith et al., 2017; Purdy and Jones, 2012; Turnhout et al., 2020). Knowledge co-production processes have been critiqued for reinforcing power imbalances and marginalizing those with alternative perspectives while sanctioned or legitimized by the participatory process. (Brenkert-Smith et al., 2017; Klenk and Meehan, 2015; Turnhout et al., 2020; Wyborn et al., 2019)

Boundaries construct, maintain, and reinforce shared understandings (Osaka et al., 2021; Wyborn, 2015). Boundary work refers to processes by which actors demarcate, i.e., draw a boundary around, what constitutes salient, credible, and legitimate knowledge (Bremer and Meisch, 2017; Gieryn, 1995). When influential actors set the boundaries of an issue, it is within those boundaries that decision-makers then diagnose the causes of a problem and design solutions (Entman, 1993; Halfman, 2019). However, boundaries can also exclude ideas from entering the debate (Jerneck and Olsson, 2011; Langley et al., 2019). This may lead actors with different understandings to avoid the process (Carboni et al., 2017; Wondolleck et al., 2003) or engage in a more limited manner (Paveglio et al., 2015; Purdy and Jones, 2012). Few studies have directly investigated how differences in

the way actors demarcate the credibility, legitimacy, and salience of their understandings shape collaborative knowledge-making (see Daly, 2016; Muñoz-Erickson, 2014; Wyborn, 2015 for notable exceptions).

Boundary management is an intentional process to facilitate knowledge sharing and integration among actors with multiple understandings by blurring, dissolving, or simply recognizing knowledge boundaries (Langley et al., 2019; Jerneck and Olsson, 2011). Boundary management is a vital component of knowledge co-production (Bandola-Gill et al., 2023; Daly, 2016; Gustafsson and Lidskog, 2018). Recognizing the boundaries that demarcate different understandings may reveal underlying bias, elevate marginalized agendas, challenge assumptions, prompt reflexivity, foster learning and mutual respect, and facilitate a greater capacity for change (Caniglia et al., 2021; Chambers et al., 2022). However, practice lacks a pragmatic approach to reveal implicit boundaries.

3. Representing shared understandings through social narratives

“Dominant narratives are not called stories. They are called reality.”
(MacKinnon, 1996, pg. 235)

Narratives may function as a meaningful way to portray actors' understandings. Central to portraying understanding is the concept of framing, a dynamic interactive cognitive process by which actors sift through complex and ambiguous information and determine which elements of a particular problem are most important (Nisbet 2009 as cited by Yung et al., 2022). There are three processes at play in framing: naming, selecting, and storytelling (van Hulst and Yanow, 2016). Narratives represent stories that weave together selected features (Schön and Rein, 1995) to render a situation more coherent and graspable (Fischer, 2003). Actors construct narratives to give meaning to or interpret their experiences and knowledge (Polkinghorne, 1995) or to suggest what is happening and what should be done (Fischer, 2003). Narratives represent deep and nuanced characterizations that map actors' frames onto linguistic tokens or discourse (Creswell, 2007; Dupraw, 2018; Fischer, 2003; Plastina, 2022). This makes them an important tool for characterizing multiple understandings of complex sustainability challenges (Brenkert-Smith et al., 2017; Innes and Booher, 2010; Paveglio, 2021). Furthermore, narratives may be especially useful for bringing multiple understandings into collaborative risk management practices, as narratives are more likely than scientific facts to encourage action-taking (Morris et al., 2019, as cited by Plastina, 2022).

Stories both shape and are shaped by the social groups they emerge from (Fischer, 2003). Narratives serve to give cohesion to shared beliefs and to transmit basic values (Fischer, 2003). As narratives multiply and transform in the social arena, i.e., are publicly told and retold, they evolve into social narratives, capturing shared experiences, understandings, and interpretations of phenomena (Shenhav, 2015). Social narratives reveal the situated socio-political framings of the situation constructed by interpretive communities (van Hulst and Yanow, 2016). Influential actors such as public officials, heads of organizations, senior scientists, and actors with large social media following play an important role in communicating and multiplying narratives, i.e., shaping social narratives. While social narratives help reveal the types of understandings that are shared - both overlapped across and circulating among social groups, it may be unproductive and overly simplistic to typecast any one community as beholden to a particular social narrative.

Frame analysis has served as an important method for investigating natural resource or sustainability challenges (Gray, 2004; Jerneck and Olsson, 2011; Pregernig, 2014; Walker et al., 2020), especially intractable challenges (Asah et al., 2012; Lewicki et al., 2003; Shmueli et al., 2006), where substantial differences in frames make conflicts “highly resistant to resolution by appeal to evidence, research, or reasoned argument” (Schön and Rein, 1995, p. xi). Frame analysis can reveal the underlying frames actors use to interpret the causes, consequences, and

solutions to challenges (Nisbet 2009 as cited by Yung et al., 2022). Frame analyses are also useful for understanding dynamic power-sensitive issues (Turnhout et al., 2020). Narrative analysis, as a genre of frame analysis, has been used to provide a schema of different understandings of how actors make sense of conflicts and identify many truths rather than finding one generalizable truth (Feldman et al., 2004; Hunter, 2010). By analyzing the language actors use to defend or justify their ideas, narrative analysis can reveal implicit boundaries actors use to demarcate what counts as knowledge (Rein and Schön, 1996). Analyzing patterns of justifications associated with different social narratives may reveal differences in the implicit boundaries associated with diverse understandings.

4. Social narratives of western wildfire challenges

Actors’ understandings of wildfire challenges, including their risk perception and prioritization of solutions to mitigate risk, are heavily influenced by their social narratives (Brenkert-Smith et al., 2019; Champ et al., 2012; Crow et al., 2017; Yung et al., 2022). Historically, social narratives of western wildfires have ranged from spiritual traditional fire narratives (Lake et al., 2017) to militaristic narratives focused on combating fires (Butler, 2009). After a series of recent fires, wildfire narratives have taken a larger role in social media (Ingalsbee, 2017) and academic research (Crow et al., 2017; Hall et al., 2015; Morehouse and Sonnett, 2010; Moskwa et al., 2018). Research characterizing wildfire narratives suggests a growing convergence toward a more unified, cohesive, co-produced narrative emphasizing landscape-scale ecological fire restoration (Brenkert-Smith et al., 2017; Goldstein and Butler, 2010). This narrative has been extensively reproduced, shaping the assumptions and expectations of the National Cohesive Strategy (Wildland Fire Leadership Council, 2014), the USDA Forest Service (Brenkert-Smith et al., 2017), the Fire Learning Networks (Goldstein and Butler, 2009), and, at a local level, Community Wildfire Protection Plans (Abrams et al., 2016; Brummel et al., 2010). Despite its prominence, wildfire risk management actions aligned with this narrative - such as, for example, prescribed burning and managed wildfires, have been slow to take root (Brenkert-Smith et al., 2017; McWethy et al., 2019; Yung et al., 2022). Burns and Cheng (2007) caution against the assumption that there is agreement on wildfire problem frames. Competing wildfire narratives hold dichotomous or conflicting perspectives about the role of fire on the landscape, posing a challenge to advancing collective cross-boundary action (Brenkert-Smith et al., 2019; Butler, 2009; Carroll et al., 2006; Crow et al., 2017; Edwards and Gill, 2016; Ganey et al., 2017; Morehouse and Sonnett, 2010). Proposed mitigation projects often struggle to advance proposals that meet the needs of groups with competing narratives (Mylek and Schirmer, 2020). While social narratives do not drive or predict wildfire conditions, identifying and characterizing social narratives can inform processes for improved representation and deliberation in risk management practices. However, while substantial attention has been paid to the role of wildfire narratives, there have been no systematic analyses that have identified and characterized them.

5. Methods

We undertook a social narrative analysis which adapts a traditional frame analysis by paying attention to shared and common storylines that are repeated and circulated among groups of actors (Shenhav, 2015). We empirically investigated three research questions: (1) what are the social narratives of western wildfire challenges? (2) what are the important distinctions between influential actors’ social narratives? and (3) how do influential actors demarcate the boundaries of credible, salient, and legitimate social narratives regarding western wildfire challenge? As a qualitative inductive approach, emergent themes further shaped the questions we investigate in this manuscript (Thornberg and Charmaz, 2014).

5.1. Data collection

5.1.1. Study population, sampling, and recruitment

This research examines fires in forested landscapes in western US states. We elicited perspectives from influential actors working on western wildfires who are recognized as experts in their field and engaged in related public discourse at a regional or national level. A purposive sampling strategy was utilized to maximize the saliency and diversity of perspectives (Schreier, 2018). We first sampled individuals from our research team’s expert knowledge (Patton, 2014) and internet searches of influential actors working on wildfire topics in western forestlands. We then used snowball sampling, identifying recommendations based on the interviewee’s first-hand knowledge (Parker et al., 2019). Selection further emphasized actors who could speak to voices that are potentially marginalized or excluded (Kristensen and Ravn, 2015). This included actors who work in the energy sector, emergency management, and community housing. Our sample included practitioners and scholars representing diverse geographies, sectors, scales, and topical areas (Table 1). We recruited actors through personalized email requests for interviews (69 % positive response rate).

5.1.2. Interviews

The research team conducted sixty interviews over the phone or video conferencing between August 2021 and August 2022. In both data collection and analysis, we tried to treat all participants equally, regardless of our agreement with their views, i.e., symmetry (Fischer, 2000) and emphasized their point of view, i.e., empathetic understanding (McIlroy-Young et al., 2021). Interviews were semi-structured (Patton, 2014), guiding participants to discuss (1) the causes, (2) consequences or impacts, and (3) potential solutions to western wildfire challenges (Appendix 1). Interviewers took detailed notes and recorded the conversation. NVivo qualitative analysis software (QSR International Pty Ltd, 2022) was used to conduct a content analysis of transcribed notes and recordings (Saldanña, 2008).

Table 1
Interviewee characteristics.

a) Professional sector and scale				
	Local	State	Regional	National
Government	1	7	5	6
NGO	4	1	8	10
Tribal	1	0	1	1
Private	1	3	1	5
Academia	0	6	6	4
b) Topics of expertise				
(n)	Topic	(n)	Topic (continued)	
16	Natural resource management & watershed protection	6	Timber industry & silviculture	
14	Fire ecology, restoration, & forest health	6	Traditional fire culture	
12	Climate change, atmosphere, & carbon	5	Community development & planning	
12	Conservation & wilderness	5	Hazards & risk management	
11	Environmental law & policy	4	Environmental justice & equity	
10	Built environment, land use codes, & utilities	4	Insurance	
10	Business innovation & partnerships	4	Public health	
10	Rural livelihoods & economic sustainability	4	Firefighting and incident command (inc. suppression)	
10	Computational models & assessment	3	Education, learning & communication	
10	Fuels management (inc. Rx fire & workforce development)	2	Cultural assets, parks & recreation	
9	Collaboration & conflict resolution			

5.2. Data analysis

We analyzed individual interview data to construct social narratives in four steps: (1) capturing individual narratives, (2) identifying repeated patterns, (3) clustering patterns, and (4) refining social narratives (Fig. 1). Our process was iterative, collecting additional data after each round of analysis (Creswell, 2007).

To capture individual narratives, we summarized participants' interviews in analytical memos. While social narratives represent shared understandings, individual narratives are unique to each actor. According to the interview guide, notes and transcripts were used to identify both how the problem is represented (i.e., diagnostic frames) and what should be done in response (i.e., prognostic frames) (Benford and Snow 2000 as cited by van Hulst and Yanow, 2016). Relationships between causes, consequences, and solutions were illustrated in conceptual models (Fig. 2). We further conducted an inductive analysis to identify emergent themes that best captured each participant's response (Ryan and Bernard, 2003). Lastly, axial coding was utilized to detect repeated patterns of relationships between themes (Glaser and Strauss, 1967). We identified four categories of repeated patterns that distinguished between participants' individual narratives:

1. **Strategies:** patterns in actors' rationale for relating solutions, causes, and consequences (Fig. 2).
2. **Scales:** different spatial and temporal scales (e.g., local short-term strategies or place-based generational knowledge).
3. **Frameworks:** participants' selection of knowledge, or activation of knowledge schema based on prior expectations about the situation (e.g., economic lens or traditional ecological knowledge).
4. **Language:** distinct terms or phrases actors used (e.g., cultural burning) representing communities of practice (Lave and Wenger, 1991; Ryan and Bernard, 2003).

We looked across transcripts and notes to identify examples of these patterns in individual narratives (Polkinghorne, 1995; Shenhav, 2015). Employing a constant comparison method (Glaser and Strauss, 1967), we iteratively compared, contrasted, and distilled emergent themes, patterns, and categories across individual narratives. Analytical memos characterized important details for each pattern of themes across the four categories (Creswell, 2007).

To construct the social narratives, we wove together patterns of strategies, scales, frameworks, and language. The highly iterative process involved repeatedly checking back against original transcripts and notes to test and reconfigure patterns across a set of social narratives (Wang, 2017). In line with narrative theory, we clustered patterns that make sense together, i.e., coherent, graspable (Schön and Rein, 1995), and internally consistent (Riessman, 2016). To the best of our ability, narratives reflect the language and meaning as portrayed by subjects, sustaining the authenticity of the told experience and respecting all ideas without judgment (Fischer, 2003). We focused on writing narratives that are illustrative, concise, and engaging to readers (Kim, 2019). Furthermore, we distinguished social narratives that are minimally overlapping or redundant (Fischer, 2003).

In addition to repeated interrogation of the interview data, we refined and finalized the selection and characterization of narratives based on feedback from multiple sources. We triangulated the social narratives against supplementary materials (e.g., publications, podcasts, conferences) for consistency and breadth (Yin, 2011). These materials helped confirm the inclusive representation of the full breadth of narratives (Meppem and Bourke, 1999), informing our decision about saturation. Drafts of the social narratives were shared with interview participants for their input. Finally, to ensure that narratives resonated with real-world experience, (Creswell, 2007; Fischer, 2003; Herrick, 2004) we elicited comments from six experts from boundary-spanning organizations who could speak to various facets of western wildfire challenges (Davis et al., 2021; Morgan, 2014). Once finalized, the analyst constructed social narratives became our unit of analysis (Ryan and Bernard, 2003). The lead author went through and recoded all relevant data (e.g., notes, transcripts) according to the social narratives.

Lastly, we sought to reveal the boundaries influential actors use to demarcate, justify, or explain multiple understandings. First, the lead author characterized and segmented justification statements across the social narratives (e.g., "there are multiple studies showing this" or "the right thing to do.") Segmented statements were grouped according to similar criteria (e.g., justifications based on rights, morality, logic, trust, pragmatism). Criteria were then compared and divided according to the extent to which they focused on credibility – believable, authoritative, and trusted information; legitimacy – fair information-producing process that considers the appropriate values, concerns, and perspectives of different or saliency – information that is relevant to decision-making

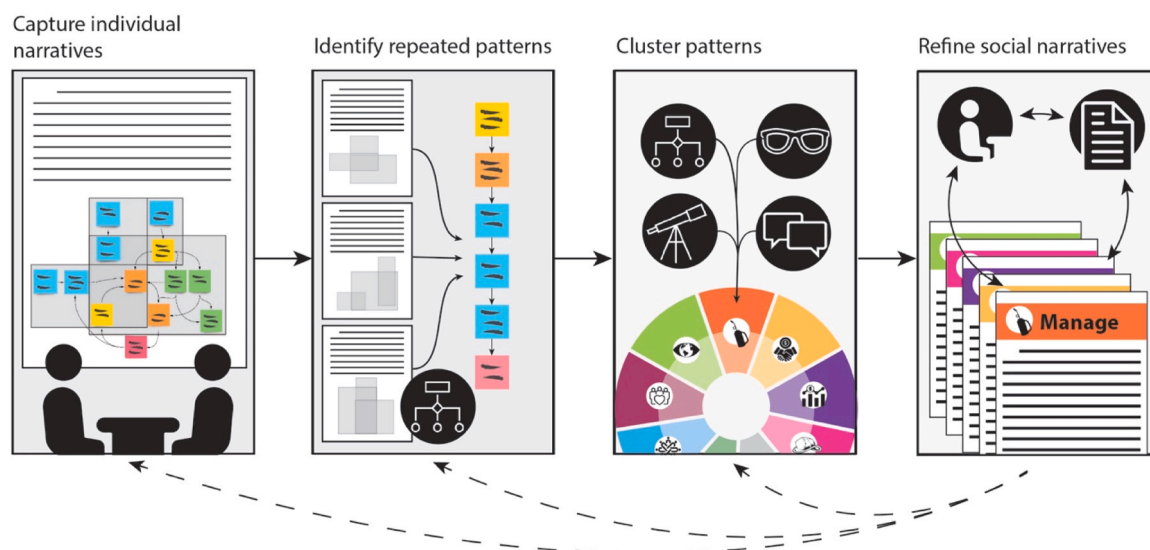


Fig. 1. Constructing social narratives. First, participant interview transcripts and notes are analyzed to capture individual narratives. Second, patterns of relationships between themes are identified across individual narratives. Third, internally consistent sets of patterns in strategies, scales, frames, and language are clustered to identify distinct social narratives. Finally, social narratives are reviewed and refined through member check-in, expert elicitation, and triangulation with supplementary materials.

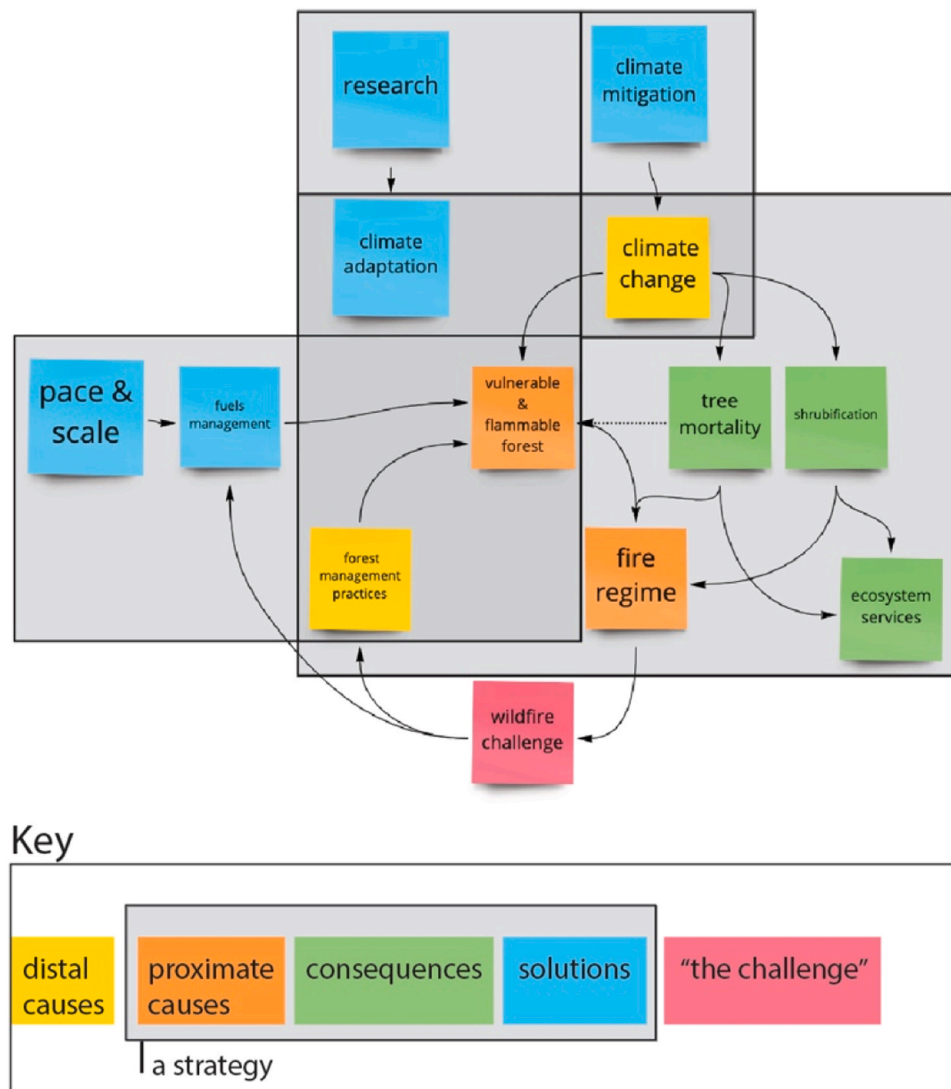


Fig. 2. Conceptual model. Conceptual models illustrate expressed relationships (arrows) between distal and primal causes of, consequences of, and solutions to wildfire challenges. Emergent strategies (bound by rectangular gray frame) capture patterns of connections between causes, consequences, and solutions heard across multiple subjects.

bodies or publics (Cash et al., 2002).

6. Results

We report on the social narratives of western wildfire challenges of sixty influential actors, organized by our three research questions. Social narratives that emerged from the interviews are in upper case and single quotes (e.g., 'Manage'), and direct quotes from participants are in double quotes.

6.1. Identifying a representative breadth of social narratives

Nine social narratives capture differences in the shared experiences and interpretations of western wildfire challenges among our interviewed participants. While this research focused on differences, there was evidence of common ground across the interviews. Everyone we spoke with recognized that western wildfires threaten social and ecological values and that society must act now to reduce impacts. There was broad agreement among all participants that conditions on the ground have changed, requiring new strategies to manage wildfire risks. Lastly, wildfire challenges were ubiquitously seen as complex, with solutions needing to match that complexity. Beyond these commonalities,

the nine social narratives are:

- 'Manage' emphasizing wildfire challenges as primarily a fuels management issue,
- 'Work' focusing on the financial sustainability of timber management to reduce risks,
- 'Market' employing an economic lens to highlight the role of insurance in signaling risk,
- 'Control' prioritizing efficiency and safety to protect against fire hazards,
- 'Regulate' pointing to the role of where and how people build to manage risks,
- 'Conserve' prioritizing forest health and restoration to reduce wildfire risks,
- 'Revitalize' emphasizing our relationship with fire, leveraging traditional fire culture,
- 'Justice' stressing the uneven distribution of wildfire burden on vulnerable populations and marginalized communities, and
- 'Adapt' underscoring the criticality of climate change as a key driver shaping future wildfire management pathways.

The rhetoric used to describe challenges further captured

participants' depictions of social narratives. We referenced direct quotes from interview participants to succinctly depict each social narrative (Table 2). For example, for 'Manage,' the statement "we know what to do, and we just need that partnership on the ground to get it done" captures its emphasis on scientific consensus and the need for immediate action. Alternatively, for 'Justice,' the statement "We must address the fact that these kinds of disasters have the most dire impacts on communities already most affected by policy inaction" illustrates its focus on the inequitable impacts on human communities. Linguistic differences were further supported by keywords representing the most frequently used unique or nonoverlapping words identified in the segmented transcripts (Table 2). Paraphrased summaries of each social narrative are provided in Appendix 2. It is important to distinguish between the names we labeled each social narrative with and the diversity of strategies with similar names (e.g., adaptation strategies are not limited to 'Adapt', and management strategies are not limited to 'Manage').

While social narratives are unique, many share perspectives that are complementary (different perspectives that are well aligned) and overlapping (a view shared by two or more social narratives). For example, 'Manage' and 'Work' both prioritize removing fuels from forests to reduce wildfire risks; however, where 'Work' focuses on monetizing fuel management to support rural livelihoods, 'Manage' focuses on landscape-scale treatments on public lands. Similarly, 'Manage' and 'Adapt' both emphasize science-based technical expertise. However, while 'Adapt' highlights the uncertainty in scientific understandings, 'Manage' emphasizes scientific consensus and certainty. Alternatively,

Table 2

Summaries of the social narratives. Key quotes reflect verbatim quotes from interview participants that capture the main idea or storyline in each social narrative. Keywords represent the most commonly used terms and phrases that are unique (nonoverlapping) to each social narrative.

Social narrative	Key quote	Keywords
Manage	"We know what to do, and we just need that partnership on the ground to get it done."	fuels, pace and scale, scientific consensus, risk management strategy, and living with fire
Work	"The west doesn't have a fire problem, it has a tree problem, and our solution is getting the trees out."	timber, products, biomass, sustainable economies, rural communities, jobs, and regulations
Market	"If we let the market do its magic, residents would bear the true risk they impose, and wildfire risks would go down."	insurance, risk, price, models, accurate, and burden
Control	"The cost of suppressing fires when they are small is much smaller and less risky than letting them get big and then trying to suppress them."	suppression, emergency, line, incident command, firefighters, ignitions, problem, and attack
Regulate	"If suddenly I was made king of the world, one of the first things I'd do is say 'no more houses in the WUI'"	harm's way, code, plans, standards, hardening, and utilities
Conserve	"There is a risk of using fuel reductions as an excuse for forest exploitation. Fuels management that needs to make a buck for somebody is likely to be counterproductive"	logging, ecology, restoring, intensity, protecting, and species
Revitalize	"All of us have a role, and all of us can revitalize our productive relationship with fire."	cultural, place, relationship, indigenous, process, stories, stewardship, learning, and generation
Justice	"We must address the fact that these kinds of disasters have the most dire impacts on communities already most affected by policy inaction."	equity, social, community, smoke, health, impacts
Adapt	"We literally have to be thinking about ways to adapt the landscape in ways that there may not be any analog for."	future, transitions, expectations, existential, and uncertainty

some social narratives hold perspectives that are more divergent or conflicting. For example, 'Revitalize' emphasizes the importance of increasing the role of fire on the landscape and democratizing fire management. In contrast, 'Control' emphasizes stopping fires while they are still small and investing in firefighter workforces. While 'Manage' emphasizes active management in forested lands, 'Conserve' and 'Regulate' emphasize little to no interventions in the forest, instead emphasizing interventions in adjacent communities. We systematically compared each pair of social narratives to look for complementary and conflicting differences. We organized the nine social narratives around a wheel, placing narratives with important complements adjacent to one another and more divergent ones across from one another (Fig. 3).

6.2. Characterizing distinctions between the social narratives

We report on four themes along which the social narratives differed: strategies, scales, frameworks, and language. Although this section focuses on characterizing distinctions between the social narratives, there is no reason wildfire risk management cannot integrate strategies associated with multiple social narratives. Furthermore, it is important to reiterate that individual narratives include perspectives associated with multiple social narratives.

6.2.1. Strategies

Strategies related between solutions, causes, and consequences. They reflected actors' logic or rationale for how their ideas will achieve a specific aim or address an underlying issue (Table 3). For example, 'Regulate' solutions emphasized home hardening, managed retreat, defensible space, and reforming utility infrastructure to change where and how actors develop. These solutions address exposure of developments within high-risk fire zones (cause), the loss of community infrastructure assets, and the costs of building back (consequences).

6.2.2. Scales

In discussing strategies, actors placed emphasis on different time periods and spatial extents at which they interpret western wildfire challenges. We conducted a qualitative thematic analysis looking for references to terms and phrases associated with different spatial and temporal scales. For example, statements associated with 'Market' were oriented around annual state-level practices, as captured by the statement: "If you're an insurance broker and you're selling this policy, the premium is based on the risk this year. Efforts outside this year are harder to quantify." Alternatively, statements associated with 'Regulate' were oriented around multi-year community-level practices, as characterized by the statement: "Our long-term strategy will need to be ongoing for decades, and it needs to be scaled down to the community level to be effective." Differences in spatial and temporal scales characterized differences in actors' priorities or emphases. For example, some actors contrasted the longer-term and landscape-scale planning aligned with the 'Manage' narrative to the shorter-term response of 'Control,' e.g., "There's a 24-hour mentality on a fire that's going to be on a landscape for two months." To visualize some of these differences, we mapped references to spatial and temporal scales in association with each social narrative onto a Stommel diagram (Vance and Doel, 2010) (Fig. 4).

6.2.3. Frameworks

Participants' emphasis on different knowledge and ideas suggested the frameworks they used to characterize western wildfire challenges. Differences in frameworks corresponded with differences in planning approaches, expectations regarding outcomes, and differences in the success of different processes (Burns and Cheng, 2007; Williams et al., 2012). Social narratives were associated with different frameworks. For example, 'Adapt' structured western wildfire challenges in terms of future changes while 'Conserve' structured challenges in relation to past events; 'Regulate' structured challenges in more technical terms, while



Fig. 3. Social narratives of western wildfire challenges. This wheel illustrates the relationship between nine social narratives of western wildfire challenges. Narratives that are adjacent are relatively more similar or complementary while narratives that are on opposing sides are more divergent or conflicting.

'Justice' structured challenges in terms of social justice and equity; 'Manage' and 'Control' structured challenges through a more paternalistic view of nature, while 'Revitalize' emphasized a more synergistic relationship to nature. Multiple social narratives used the same framework. For example, both 'Work' and 'Market' structured western wildfires as economic challenges, the former emphasizing market-based solutions like supporting the timber industry, while the latter setting insurance premiums. Social narratives often referenced multiple frameworks. For example, while 'Adapt' structured western wildfire challenges as future-oriented, it also structured western wildfires as ecological challenges focusing on ecosystem functions, intrinsic values, and more extended returns on investments.

6.2.4. Language

The unique language actors used helped distinguish between social narratives. Indigenous or emic terms and phrases helped reveal taken-for-granted assumptions and different communities of practice (Butler, 2009; McIlroy-Young et al., 2021). For example, "pace and scale," "no analog conditions," "culture of fear," and "let it burn" came up as common linguistic patterns that helped distinguish between the social narratives. Unique terms may reveal the elements multiplied by social narratives that connect actors to one another. Sometimes, participants explicitly called out language differences between the social narratives. These terms may portray parts of their understanding that are omitted, simplified, or misunderstood. For example, a handful of actors emphasized the oversimplification between the terms "cultural burning" and "prescribed burning," distinguishing the diversity of methods,

objectives, and spiritual intentions involved in cultural burning. Differences in how actors depicted the same element also distinguished between social narratives, e.g., "They call it fuels, I call it vegetation because its habitat, not just fuels." Similarly, 'Manage' labeled unplanned ignitions as "managed wildfire," described as a practice essential to reducing fuels at the scale necessary to achieve risk reduction objectives. In contrast, 'Control' labeled unplanned ignitions as "letting fires burn," described as a risky practice that misappropriates federal funds.

6.3. Describing differences in the boundaries of credible, salient, and legitimate social narratives

To understand the boundaries around social narratives, we analyzed the statements participants used to justify their evidence and ideas. We found that each social narrative corresponds with a different set of justifications. A diverse set of criteria justified what counts as knowledge according to the categories of credibility, legitimacy from experiential knowledge to empirical evidence, or vetted information (Table 4). For example, 'Manage' justified credibility by referencing scientific consensus and technical solutions, while 'Revitalize' justified credibility through generational knowledge. The generational value of knowledge reflected in Revitalize was well captured by the statement: "The average five-year-old Karuk child knew more about fire than 99.9 % of the current population of the United States. They were taught by their elders from a very young age how to use fire." Alternatively, 'Control' emphasized saliency through personal experience and professional

Table 3
Causes, consequences, and solutions associated with different social narratives.

	Causes	Consequences (focus)	Solutions to wildfire challenges
Manage	Suppression; forest management; fuel-dense forestlands	Forestlands; size, frequency, and magnitude of wildfires; cost of suppression	Increase pace and scale, e.g., thinning, Rx, managed wildfire; change incentive structure; models and risk assessments; workforce development; research and funding; outreach and communication; social sanction
Work	Environmental regulations; loss of timber management	Financial viability; rural communities; livelihoods	Streamline regulations; connect jobs to actions; innovations and partnerships; biomass utilization; access roads and fuel breaks
Market	Poorly understood change in risk, insurance reform so premiums don't match risk, risk too expensive	Cost to taxpayers; assets (e.g., value in home); insurance (i. e., exiting); predictability for homeowners and insurers	Allow premiums to reflect risk signal; minimize insurance regulation; investments in accurate catastrophic models; cross-state model reviews; cost-benefit analysis of household risk
Control	Human ignitions, managed wildfire practices, lack of swift initial attack, letting wildfires get out of control	Health and safety; firefighters; assets; communities; cost of fire management	Mitigation strategies Direct attack; quick initial attack; strategic use of trained workforce; investments in workforce; evacuations; better forest management practices
Regulate	Development in harm's way; increase exposure and hazards; electric utilities	WUI communities; assets; cost to restore and build back; utilities and infrastructure systems	Community planning and engagement; house designs and retrofits; location of development; local plans and policy; zoning and development codes; defensible space; regulation of public utilities; managed retreat
Conserve	Extractive forestry practices	Forest health; resiliency, watershed health;	Ecological fire restoration; passive management in wilderness; forest protections; conservation; environmental regulations; managed wildfires
Revitalize	Criminalization of cultural burning and traditional fire practices	Culture; spiritual connections; traditional foods; safety; identity	Indigenous training; cultural burning, greater rights and access to manage lands; authentic collaboration
Justice	Socio-economic and environmental inequities; lack of protective measures; global patterns of environmental crises	Marginalized and vulnerable communities; unequal burden; affordable housing; smoke and public	Community planning and engagement; vulnerability and adaptation plans; better relationships; smoke management;

Table 3 (continued)

	Causes	Consequences (focus)	Solutions to wildfire challenges
Adapt	Climate signal; hotter temperatures; vapor pressure deficit; drier conditions; winds; natural ignitions	health; reliable utilities (e.g., public safety power shutoff) Fire seasons; landscape patterns (e.g., shrubification, zombie forests); more emissions; ecological instability	research on health impacts; affordable housing; standards for outdoor labor Reduce emissions (climate mitigation); adaptation; change expectations; integrated predictive models

knowledge, e.g., “boots-on-the-ground” and “end-of-a-Pulaski-with-my-kids-on-the-fire-line.”

Our findings provide numerous examples of actors reaffirming or maintaining what counts as knowledge and what knowledge may be less salient, credible, or legitimate. For example, one participant reaffirmed the boundary between science and practice, or facts and values, by remarking that – “too often solid science across a range of disciplines is conflated with feelings and partial truths that render them less than useful to further discourse.” We also heard how organizational culture and peer review can reinforce boundaries (Miller et al., 2008), e.g., “If you want to get the grant, you need to use the right language” or “Here’s a warning for when you write an article, make sure to say within the first paragraph: wildfires are getting worse because of fuels buildup due to 100 years of fire suppression, climate change, and the ever-expanding wildland urban interface.”

While we found evidence of boundary maintenance, we also found evidence that disciplinary and sector-based boundaries may be dissolving or blurring. As previously noted, while interviewees might have leaned towards one social narrative more than another, most actors were likely to identify with aspects of multiple narratives. Furthermore, participants from the same organization or actor group did not consistently align with the same social narratives, and participants combined multiple social narratives inconsistently. For example, a handful of participants combined elements of ‘Manage’ and ‘Conserve’ to emphasize a more ecological restoration of fuel management while remaining skeptical about larger-scale logging and thinning operations. Alternatively, a handful of participants combined elements of ‘Manage’ with the ‘Work’ narrative, emphasizing the important role of partnerships between federal land managers and industry to scale up fuel reductions. This may indicate that the criteria for what counts as knowledge is changing. It also confirms that individuals and groups should not be typecast into any one social narrative.

7. Discussion

Practitioners are increasingly turning to knowledge co-production to address complex sustainability challenges (Lemos et al., 2018). However, several challenges may hinder the capacity of knowledge co-production processes to recognize and legitimize multiple understandings (Jagannathan et al., 2020; Norström et al., 2020). Our findings suggest that narrative analysis may provide a pragmatic approach to address these challenges. Based on our interviews, we identified nine social narratives, characterized important distinctions between the narratives, and revealed different boundaries actors use to demarcate social narratives of western wildfire challenges. Our findings have implications for informing western wildfire risk management practices and, more broadly, complex sustainability challenges.

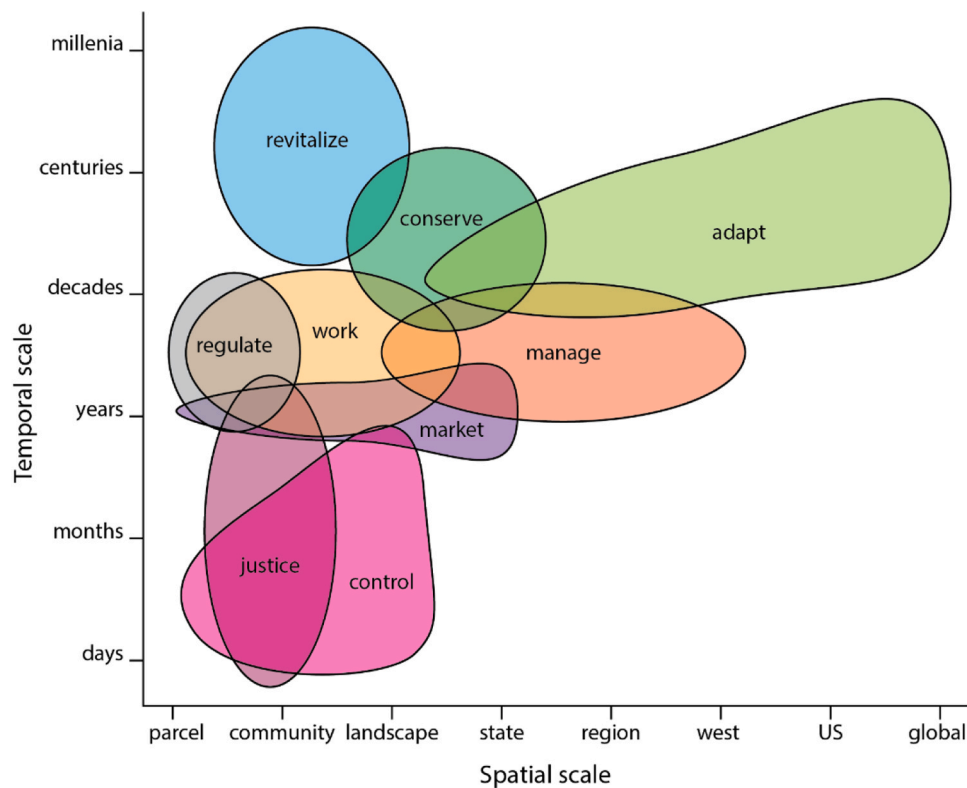


Fig. 4. Temporal and spatial scales of social narratives. This Stommel diagram illustrates the temporal and spatial scales of each of the nine social narratives of western wildfire challenges. Scales correspond with the strategic elements of each narrative, characterizing the scale at which causes, consequences, and solutions are conceptualized. Differences in the scale referenced by actors may help explain some mismatch between their understandings.

7.1. Implications for addressing western wildfire challenges

7.1.1. A representative breadth of understandings to support more inclusive and effective practices

Past research suggests that differences among actors' understandings can shape the efficacy of collaborative wildfire risk management (Abrams et al., 2016; Brenkert-Smith et al., 2017; Paveglio et al., 2018). Our findings of nine social narratives suggest there are far more understandings to be recognized in wildfire risk management practices than are currently recognized. While elements from each of the nine social narratives are noted elsewhere, this is the first systematic investigation that brings them all together. Previous characterizations have depicted differences in terms of their dichotomy (Butler, 2009; Crow et al., 2017; Ganey et al., 2017), conflict, (see Brenkert-Smith et al., 2019; Crow et al., 2017; Edwards and Gill, 2016), or unity (Brenkert-Smith et al., 2017; Butler, 2009). Our findings reveal that how actors interpret and engage with wildfire challenges is far more complex and overlapping.

Identifying a multitude of wildfire narratives suggests that now might be an especially important time to address differences in actors' understandings. This proliferation may imply a 'constitutional moment' where taken-for-granted assumptions are challenged and renegotiated (Jasanoff, 2003). The growing saliency of wildfires and political and economic investments may have catalyzed this moment (Burke et al., 2021). While some narratives have shaped management for decades, 'Revitalize,' 'Justice,' 'Market,' and 'Adapt' have likely evolved as recent wildfire catastrophes intersect with more contemporary topics such as climate change, equity, and insurance failures. These focusing events (Birkland, 2006; Crow et al., 2017) or windows of opportunity (Chapin et al., 2010; Newig et al., 2019) may signal that the time is ripe for catalyzing institutional change (Wyborn et al., 2019).

7.1.2. Explicit distinctions to inform the design of practice

Distinguishing complementary and conflicting differences between social narratives can influence the design of practice (Chambers et al., 2022; Devente et al., 2016; Hakkarainen et al., 2022). Characterizing differences doesn't eliminate them; however, being aware of differences helps knowledge co-production processes be more prepared for them (Wyborn et al., 2019). We identified four categories of themes that capture differences in social narratives: strategies, scales, frameworks, and language. These distinctions align with prior research describing tensions and friction between actors (Brenkert-Smith et al., 2017; Steelman, 2016; Williams et al., 2012). For example, social narratives may clash over prioritizations of short-term solutions versus long-term outlooks or financial versus ecological implications. Having actors identify and discuss differences before making decisions may help anticipate or prevent conflict (Lang et al., 2012). It may also inform discussions about tradeoffs (Hamilton et al., 2019).

Complementary differences emphasize positive attributes of diversity to increase efficacy, inclusion, innovation, and adaptation (Mannix and Neale, 2005). Complementary differences among actors' understandings have benefited numerous Collaborative Forest Landscape Restoration Projects (Schultz et al., 2012) and Community Wildfire Protection Plans (Abrams et al., 2016). Alternatively, conflicting understandings often stem from deep positional differences or identity challenges and can delay or derail collaborative practices (Dewulf et al., 2009; Goldstein et al., 2010; Shmueli et al., 2006). In line with prior research, we found evidence of conflicting differences where risk perception and core values are in play (Burns and Cheng, 2007; Vaughan and Seifert, 1992). "Managed wildfire" and "letting wildfire burn" are examples of linguistic differences that depict a conflict over the use of unplanned ignitions to manage wildfires. We found examples of negative characterizations and defensive posturing suggesting identity challenges between the 'Work' and 'Conserve' and, to a lesser extent, between the 'Revitalize' and 'Manage' narratives. These identity

Table 4

Knowledge demarcation. Criteria for justifying what counts as knowledge according to the categories of credibility, legitimacy, and saliency. Dots indicate whether criteria were salient within narratives. Statements are example quotes from interview participants to characterize each criterion.

Justification	Criteria	Manage	Work	Market	Control	Regulate	Conserve	Revitalize	Justice	Adapt	Statement(s)
Credible. Believable, authoritative, trusted information (Cash et. al., 2002).	Empirical	●							●	●	"based on real data" "you can throw away these data points"
	Certain	●				●					"we have consensus on the science", "100 % it would reduce the risk"
	Scientific expertise	●								●	"there's massive uncertainty", "it's really complex" "the ologists and experts", "the 'public' doesn't understand"
	Logic			●	●		●			●	"I'm who they go to" "it's clear that it hasn't worked out", "it just doesn't add up"
	Trust		●				●		●		"a long history of distrust" "an underlying suspicion" "trauma, trust, and violence of colonialism"
	Validity	●								●	"I don't see any scientific validity"
	Objective						●	●			"these things are just objectively true"
	Professional	●	●		●						"we know how to do a good operation", "doing this a long time"
	Technical									●	"faster, more agile models", "running thousands of simulations"
	Vested interests			●					●		"dramatically under measured" "they have the money, the resources, and the financial incentive ... so that's going to be as close to the true price signal of risk"
	Vetted										"there are multiple studies showing this", "credible peer-reviewed publications"
	Authentic		●							●	"giving people false hope" "there's a bit of hypocrisy"
	Generational							●			"pretense of collaboration" "intergenerational transfer of traditional knowledge", "a much longer story", "we've coexisted with fire for millennia"
Legitimate. A fair information producing process that considers the appropriate values, concerns, and perspectives of different actors (Cash et. al., 2002).	Natural						●				"some of these fires are just naturally going to be larger"
	Rights							●	●		"the right to self-determination", "not only knowledge, but rights" "who decides what the stories are", "exclusion of people"
	Insider		●	●							"firefighters tell me", "we gotta help our people"
	Just									●	"exacerbating social inequities and injustices"
	Moral				●					●	"protect the health and safety", "a very real human reason"
	Spiritual						●	●			"the right thing to do" "deep metaphysical relationship", "it's deeper. It's highly spiritual,"
	Socio-cultural					●		●			"culture, but also social system, the regalia" "it's the social aspect of it, like, I'm not going to cut down this rose bush because"

(continued on next page)

Table 4 (continued)

Justification	Criteria	Manage	Work	Market	Control	Regulate	Conserve	Revitalize	Justice	Adapt	Statement(s)
Salient. Information that is relevant to decision making bodies or publics (Cash et al., 2002).	Other ways of knowing							●	●		my husband, who has passed away, planted that." "science is important, but it's just one part of the equation", "hard to accept that there is more than one way to do this"
	Experiential	●			●						"boots-on-the-ground", "they have never been to a fire", "end-of-a-Pulaski-with-my-kids-on-the-fire-line"
	Effective		●	●							"the most effective, efficient, and economical way"
	Place-based							●	●		"localizing the relationship to fire stewardship"
	Quality		●					●	●		"indigenous means native to a place"
	Current						●			●	"what is the quality of these jobs", "who is going to bear the cost"
	Financial	●		●							"what is it going to do to the quality"
	Important		●		●	●	●	●	●		"we need models that reflect where we are now", "new normal", "20 years ago is not relevant", "not your grandfather's fire"
	Pragmatic	●	●			●					"there's no actuarial basis for what these discounts should be"
				●						●	"we cannot afford to allow fire on the landscape"
											"there's a really important conversation"
											"that's huge, a game-changer"
											"a commonsense sort of point of view", "a reasonable place"

challenges may reflect lingering resentments from historical conflicts, like the 'Timber Wars' and the criminalization of traditional fire practices (Davis and Lewicki, 2003; Tedim et al., 2021). Resolving these conflicts may require skilled mediation and longer relationship-building efforts (Gray, 2004; Santos et al., 2022; Shmueli et al., 2006).

In addition to differences, characterizing overlaps or similarities across social narratives can inform the design of collaborative practices. As exhibited, different understandings utilize different strategies, scales, frameworks, and terminology. Translation problems across different interpretations can arise and prohibit or delay collaborative action (Brugnach and Ingram, 2012). Characterizing similarities can help bridge connections between groups and overcome perceptual barriers. We identified overlaps across all social narratives, even seemingly opposing ones. For example, in terms of strategies, while 'Manage; and 'Conserve' may hold opposing ideas about active management, they both emphasize forest health and managed wildfires. 'Regulate,' attributing causality to development in harm's way,' and 'Justice,' emphasizing the impact on those communities, may have divergent understandings of the situation, however, both overlap by operating at a community and annual scale.

7.1.3. Characterizing implicit demarcation boundaries to reveal power imbalances

By analyzing different justifications of social narratives, we revealed the implicit boundaries actors use to demarcate what counts as knowledge. Our findings suggest that each social narrative is associated with a distinct set of justifications. Wildfire risk management collaborations that impose one set of assumptions about what evidence counts as

knowledge are likely to silence or alienate actors who hold a different set of assumptions. For example, a process emphasizing science-based decision-making may effectively exclude actors who resonate more with 'Work,' 'Justice,' or 'Control' narratives. The frustration felt by actors asked to engage in processes where these assumptions are taken for granted is captured in the statement, "Collaboration from the sense of the agency was we're going to listen to you, and then we're going to do whatever we want." While revealing differences in how actors delineate the boundaries of the credibility, legitimacy, and saliency of their understandings does not eliminate power imbalances, it can help facilitate how discursive power is brought into a conversation.

Wildfire challenges were ubiquitously seen as complex, with solutions needing to match that complexity. We found participants' understandings to be equally complex. This was partly evidenced by participants' individual narratives referencing multiple social narratives in diverse ways. However, we found a potential discrepancy between the complexity of actors' understandings and actors' assumptions about the simplicity or homogeneity of other actors' understandings. Statements like "they don't understand the importance of fire on the landscape" or "they start with a legitimate argument but take it into ridiculous places" exemplify some of the language actors use when oversimplifying the understandings of other actors. These characterizations perpetuate stereotypes and may distance or polarize actors. They may further overlook overlaps across actors' understandings. Actor groups or stakeholder types are not stuck with any one social narrative. Facilitating conversations where actors discuss differences and similarities in their understandings can provide opportunities for finding connections and improving collaborative capacity (Innes and Booher, 2016).

7.2. Broader implications for complex sustainability challenges

Our research suggests that analyzing social narratives fills a significant gap in knowledge co-production processes by providing a pragmatic approach to identify and characterize multiple understandings and reveal their boundaries. Narrative analysis can aid the initial exploration stage of co-production (Steger et al., 2021) by capturing both the breadth of and important distinctions between actors' understandings. This research highlights the need to engage more deeply with the complexity of actors' understandings when designing knowledge co-production processes (Fazey et al., 2013). Traditional narrative analysis generally associates each entity with one narrative (Creswell, 2007; Fischer, 2003). However, to capture a representative breadth of understandings, we suggest distinguishing individual narratives from social narratives and individuals' narratives from those of their organization or interest group. Most processes select group membership according to sector, organization, or interest-group (Reed et al., 2009), making assumptions about a correspondence between these diversity measures and understandings. Our findings challenge that assumption. Aligned with prior studies, we found that actor type does not necessarily predict perspective (Brouwers et al., 2022; Goldman et al., 2018). Demographic groups should not be associated with a singular social narrative. Today's environmentalists, private industry, and federal forest managers are heterogeneous and traverse knowledge boundaries in a multitude of ways (Jaworsky, 2016; Langley et al., 2019). We found that actors resonate with multiple social narratives, and actors from the same actor type or organization may not resonate with the same narratives. This makes sense as influential actors are exposed to diverse narratives through their multiple communities of practice and interdisciplinary collaborations.

For risk management practices to be authentically collaborative, they must recognize and legitimize multiple narratives. Opportunities to explicate and reflect on diverse storylines can help actors be more reflexive about their own partial and positioned understanding (Bixler et al., 2022; Brouwers et al., 2022; Chambers et al., 2022; Polk, 2015), enable them to consider a broader range of solutions (Gray, 2004; Yung et al., 2022), and collectively reframe or author new narratives (Goldstein and Butler, 2010; Rein and Schön, 1996). Constructing a shared understanding (Brenkert-Smith et al., 2017; Jacobson et al., 2021) or meta-frame (Norgaard, 2010; Schultz et al., 2019) that integrates across different understandings has been identified as an important approach to create common ground and move practice forward (Butler and Schultz, 2019; Cash et al., 2003; Schultz et al., 2019). However, as captured by Klenk and Meehan, the imperative to integrate multiple understandings may conceal friction, antagonism, and power, excluding important distinctions in the diversity of ways to understand complex challenges (2015). Social narrative analysis provides a mechanism to systematically capture important distinctions between understandings and refine how understandings are brought together. Rather than ignoring or minimizing multiple understandings and their underlying socio-cultural differences, explicating differences and commonalities can help re-politicize knowledge co-production (Turnhout et al., 2020).

Our focus on the demarcation of social narratives further revealed how different understandings are maintained by various types of justifications (e.g., empirical, ethical, professional). Explicitly revealing demarcation boundaries suggests ways to bring multiple understandings into conversation. Collaborative environmental governance practices increasingly emphasize using evaluation metrics to determine what evidence should count in decision-making (Fazey et al., 2014; Meadow et al., 2015; Wall et al., 2017). Our research indicates that actors with different understandings hold different ideas about what metrics to use when deciding what evidence to include. This research makes the case for engaging diverse actors in all stages of knowledge production, including determining evaluation criteria (Chakraborty et al., 2022; Dale and Armitage, 2011; Tengö et al., 2014).

7.3. Research limitations

Narratives are inherently subjective as no concrete phenomenon validates or confirms findings. These social narratives reflect common experiences and interpretations of individuals actively working in and well versed in western wildfire challenges. They do not capture the social narratives of residents or politicians, may carry little relevance as conditions on the ground and social norms shift, and may not apply to wildfire in non-forested landscapes such as grasslands. Given our non-representative sample and the qualitative nature of our semi-structured interviews, validation strategies like intercoder reliability and nonparametric statistics are unsuitable (Morse, 2017).

7.4. Future research directions

There are a few ways in which this research can be built upon. First, further exploration is needed to quantify the distribution of social narratives among the population engaged with western wildfire challenges. For example, a larger random sample and a more structured questionnaire or survey would be required to determine which narratives are more common. Second, among the influential actors we spoke to, we heard the mixing of multiple social narratives in diverse ways. Future research could examine which narratives correspond more closely (i.e., are often told together) and which are more in conflict. It could also identify the emergence of a new hybrid narrative among actors whereby the same narratives are consistently told together. Third, social narratives represent shared understandings among like-minded actors or groups. However, the extent to which social narratives align with interest groups, disciplines, or other attributes must be clarified. How might practice determine selection criteria if social narratives do not correspond with interest groups such as industry, NGOs, or academia? To include multiple understandings in practice, conveners must establish a clear connection between member selection criteria and actors' understandings (Ansell et al., 2020).

8. Conclusion

The understandings actors bring into a process affect the quality of the knowledge produced and, ultimately, the decisions informed by that knowledge (Berkes, 2009; Norström et al., 2020; Mach et al., 2020). If knowledge co-production processes promote recognizing and legitimizing multiple understandings, a critical precondition informing the design of collaborative practice should be greater transparency in determining those understandings. This study constructs social narratives to identify and characterize multiple understandings of western wildfire challenges.

This research holds significance for addressing western wildfire challenges by informing a more representative breadth of understandings, shaping process design, and revealing discursive power to improve wildfire risk management practices. Over the past year, we engaged in several working sessions where these social narratives sparked discussions within professional groups, shaping how they engage with diverse actor groups. While this research focuses on western wildfires, it is relevant to similar limitations in cross-boundary wildfire risk management collaboratives worldwide (Paton et al., 2014; Reid and Beilin, 2015).

This research addresses key scientific debates to bridge the gap between the aspirations of scholarship and the pragmatic needs of practice. It addresses the criticism that knowledge co-production is not pragmatic enough to allow it to be effective (Horcea-Milcu et al., 2022). Knowledge co-production is often depicted by the tension over how to balance scientific rigor and inclusivity (Cravens and Ardoin, 2016; Lang et al., 2012; Polk, 2015). While not resolving the tension, this research provides explicit characterizations of multiple understandings to engage in it more transparently. Furthermore, scholarship continues to problematize what it means to bring together multiple understandings,

emphasizing the drawbacks of reducing and concealing important differences (Klenk and Meehan, 2015). Social narrative analysis provides a mechanism to systematically capture important distinctions between understandings to refine the method by which understandings are brought together. Perhaps most importantly, despite its aspirations, knowledge co-production is often criticized for not addressing and even amplifying power imbalances (Brugnach and Ingram, 2012; Djenontin and Meadow, 2018; Jagannathan et al., 2020; Zurba et al., 2022). Revealing differences in actors' demarcations of what counts as knowledge makes a clear case for engaging diverse actors in determining evaluation criteria. Addressing these limitations may be especially necessary for international co-production efforts where actors with substantially diverse backgrounds, disciplines, cultures, and beliefs must come together (Rathwell et al., 2015; Roux et al., 2017; Wagner et al., 2023).

Identifying and characterizing multiple understandings is only the first step. For knowledge co-production processes to recognize and legitimize multiple understandings, collaborative teams must directly reflect on the role multiple understandings should play. Creating room for different ways of understanding the problem and solution requires actors to reflect on the complexity and biases in their own understandings (Wyborn et al., 2019). Beyond identifying and characterizing understandings, these analyst-constructed social narratives can facilitate a conversation in which actors discuss differences in understandings and reflect on the role of discursive power in process design (Rathwell et al., 2015). Recognizing and legitimizing multiple understandings in practice will further require changes to incentive structures bolstered by institutions and society, for example, through funding for transdisciplinary research (Chapman and Schott, 2020; Thompson et al., 2017) and inclusive criteria for science-based evaluations (Stelman et al., 2021). Finally, just as narratives reflect enduring and impactful framings of the situation (Paveglio et al., 2015), they also evolve and hybridize as new knowledge is produced and socio-political contexts change (Brenkert-Smith et al., 2017). The process of identifying and characterizing social narratives is, therefore, never final or definitive but instead provides a snapshot of a moment in time.

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CRediT authorship contribution statement

Alexandra Paige Fischer: Writing – review & editing, Validation, Methodology, Investigation, Funding acquisition, Conceptualization. **Michael Russo:** Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Heidi R Huber-Stearns:** Writing – review & editing, Validation, Methodology, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.envsci.2024.103824](https://doi.org/10.1016/j.envsci.2024.103824).

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